

Shrey Aryan

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 [shrey183.github.io](https://github.com/shrey183)

Research Interests

I am broadly interested in Geometric Analysis, PDEs and Optimal Transport.

Education

MIT , Cambridge, USA	2022 - 2027
PhD, Mathematics, LANDIS FELLOW, SIMONS DISSERTATION FELLOW Advisor: Prof. Tobias Colding	
ETH , Zürich, Switzerland	2020 - 2022
M.S., Mathematics, ESOP SCHOLAR Thesis: Stability of Hardy Littlewood Sobolev Inequality Advisor: Prof. Alessio Figalli	
École Polytechnique , Palaiseau, France	2017 - 2020
B.S., Mathematics and Computer Science, CLASS VALEDICTORIAN Thesis: Existence of two-solitary waves with logarithmic distance for the nonlinear Klein–Gordon equation Advisor: Prof. Yvan Martel	

Publications

1. Non-uniqueness of geodesic limits and a question of Grayson and Gage (joint with Tang-Kai Lee). [arXiv:2608.04855](#) [[math.DG](#)].
2. Calabi-Yau Conjecture for Minimal Hypersurfaces in $\mathbb{R}^4$ with bounded geometry (joint with Alexander McWeeney and Giuseppe Tinaglia). [arXiv:2608.04364](#) [[math.DG](#)].
3. A Topological Bernstein Theorem for Minimal Hypersurfaces in $\mathbb{R}^4$ confined in space (joint with Alexander McWeeney). [arXiv:2607.05755](#) [[math.DG](#)].
4. Spectral Obstructions to Contracting Transport Maps in Curved Spaces. [arXiv:2605.24705](#) [[math.DG](#)].
5. On the Calabi–Yau Conjectures for Minimal Hypersurfaces in Higher Dimensions (joint with Alexander McWeeney). [arXiv:2602.16048](#) [[math.DG](#)].
6. Continuous in time bubbling and Soliton Resolution for Non-negative Solutions of the Energy-Critical Heat Flow. [arXiv:2512.18840](#) [[math.AP](#)].
7. Entropic Selection Principle for Monge’s Optimal Transport (joint with Promit Ghosal). [arXiv:2502.16370](#) [[math.PR](#)].

8. Soliton Resolution for the energy-critical nonlinear heat equation in the radial case . To appear in Analysis and PDE, [arXiv:2405.06005 \[math.AP\]](#).
9. Free energy minimizers with radial densities: classification and quantitative stability (joint with Lauro Silini). [arXiv:2412.03997 \[math.AP\]](#)
10. Stability of Wu's logarithmic Sobolev inequality via the Poisson-Föllmer process (joint with Yair Shenfeld and Pablo López Rivera). To appear in Electronic Communications in Probability.
11. Concavity for elliptic and parabolic equations in complex projective space (joint with Mike Law). [arXiv:2403.16783 \[math.AP\]](#). To appear in Calc. Var. Partial Differential Equations.
12. Trend to equilibrium for flows with random diffusion (joint with Matthew Rosenzweig and Gigliola Staffilani), International Mathematics Research Notices (2024), [arXiv:2307.03147 \[math.AP\]](#).
13. Stability of Hardy Littlewood Sobolev Inequality under Bubbling, Calc. Var. Partial Differential Equations 62 (2023), no.8, Paper No. 223, [arXiv:2109.12610 \[math.AP\]](#).
14. Existence of two-solitary waves with logarithmic distance for the nonlinear Klein-Gordon equation, Commun. Contemp. Math. 24, 2050091 (2020), [arXiv:2010.04852 \[math.AP\]](#).

Teaching and Mentoring

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|---|------|
| 1. UROP Mentor, MIT | 2026 |
| 2. Recitation leader for Multivariable Calculus, MIT | 2025 |
| 3. TA for a Summer School in Optimal Transport at SLMATH (formerly known as MSRI) | 2025 |
| 4. Research Summer Institute Mentor (RSI), MIT | 2024 |
| 5. Directed Reading Program Mentor, MIT | 2023 |

Talks

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| 1. Geometric Analysis Seminar, Brown | 2026 |
| 2. AMS Sectional (Metric, Analytic, and Classical Methods in Differential Geometry), Boston | 2026 |
| 3. AMS Sectional (Geometric Flows), Boston | 2026 |
| 4. Analysis Seminar, Yale | 2026 |
| 5. Probability Seminar, MIT | 2026 |
| 6. Analysis Seminar, Columbia | 2025 |
| 7. Colloquium Talk, Indian Statistical Institute | 2025 |
| 8. Colloquium Talk, KSU | 2025 |
| 9. Graduate Lecture series in Analysis and PDEs, Brown | 2025 |
| 10. Harmonic Analysis and Differential Equations Seminar, UC Berkeley | 2024 |
| 11. Short Contributed Talk at the Nonlinear PDEs Summer School, UC Berkeley | 2024 |

Referee Work

Crelle and Mathematische Annalen.